

Root Cause Analysis Report of Defective Batches in Food Processing

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Introduction

Prepared for FreshHarvest Foods Ltd., this report delves into the root causes behind defective batches in the production line. The analysis harnesses historical production, quality, and sensor data to expose key weaknesses spanning ingredient quality, supplier variations, and operational factors. It aims to guide improvements ensuring safer, higher-quality food products and optimizing manufacturing efficiency.

The insights presented will assist quality control managers and stakeholders in implementing targeted corrective actions to reduce defects and improve overall product standards.



Project Objective and Production Process Overview



Identify root causes of defects in batches using comprehensive data analysis covering production, quality checks, and sensor readings.



High-Level Process Fault Areas Linked to Defects

1

Supplier and Raw Materials

Variations in ingredient quality grades and sources impact defect rates significantly.

2

Processing Equipment

Machine parameters including duration, line number, temperature and pressure influence batch outcomes.

3

Operator Influence

Operator handling differs across products and shifts, affecting defect frequency.

4

Shift and Production Timing

Shift schedules and batch durations create variability in product quality.

The confluence of these key areas forms the foundation for further detailed observations documented in subsequent slides.

Defective Batch Statistics and Acceptable Quality Level

5000
Total Batches

562
Defective Batches

11.23%
Defective Rate

2.5%
Above AQL Standard

3.9%
Defective Material Rate

The analysis of 5,000 batches reveals a defective batch rate of 11.23%, notably exceeding the acceptable quality level (AQL) standard of 2.5%. Furthermore, defective material rate stands at 3.96%, highlighting material quality challenges. These metrics indicate a pressing requirement for intervention across multiple production stages to improve overall batch quality.

Observation 1: Product and Ingredient Defect Rates

Product Defects

MEAL002 exhibits the highest defective batch rate at 4.42%, followed by VEG003 at 4.11% and SOUP001 at 3.35%. The spread indicates product-specific vulnerabilities despite shared ingredients.

Ingredient Defects

- Carrots: 4.37%
- Broccoli & Peas: 4.02%
- Rice: 3.82%
- Chicken: 3.56%

Ingredient quality is a critical root cause impacting batch outcomes due to variable defect rates.

Observation 2: Ingredient Quality Grades and Defects

1 Grade C Quality

Highest defective rate at 4.62%, indicating low-grade materials are a significant contributor to defects.

2 Grade A Quality

Moderate defect rate of 3.95%, showing even high-grade materials can present issues impacting quality.

3 Grade B Quality

Lowest defect rate at 3.76%, suggesting mid-grade materials currently perform best across batches.

The link between ingredient quality grade and defective batches necessitates stringent quality control and supplier engagement to enhance material standards.



Observation 3: Supplier Impact on Material Quality

Uniform Grade Issues Across Suppliers

All suppliers provide ingredients with similar grade defect patterns: Grade C highest defects, followed by Grade A and B.

Supplier Quality Control

Supplier practices directly influence batch quality and require collaborative improvement efforts.

External Addressing

Supplier-side quality challenges are root causes that need external solutions beyond plant operations.

Strengthening supplier partnerships and intervention strategies will be vital for durable quality improvements in raw materials.



Observation 4: Shift and Line Number Defect Rates

Shift Defects

- Evening shift highest at 4.20%
- Morning shift at 4.13%
- Night shift lowest at 3.54%

Shift timing influences defect rates with potential operational or resource allocation factors.

Line Defects

- Line 2 highest defective rate at 4.63%
- Line 4 at 4.37%
- Line 3 at 3.86%
- Line 1 at 3.80%

Certain production lines are associated with higher defect frequencies, indicating possible mechanical or process issues.

Observation 5: Production Duration and Defect Rates

1 High Defect Timing

Batches lasting 3 hours show the highest defective rate at 4.89%, followed by 6 hours at 3.94%.

2 Lower Defect Timing

2-hour batches have 3.83% defects, while 5-hour durations see 3.36%, indicating optimal production times may exist.

3 Machine Specification Focus

Machine operational parameters and batch durations must be reviewed and optimized to minimize defects.

Addressing mechanical and scheduling controls will be pivotal in reducing batch defects attributable to production timing.



Recommendations for Quality Improvement

1

Ingredients and Grades

- Reduce Grade C material usage; prioritize Grades A and B
- Collaborate with suppliers to boost raw material quality
- Enhance raw material inspections focusing on carrots, broccoli, and peas

2

Production Improvements

- Repair and maintain Line 2 and Line 4 equipment promptly
- Control batch production duration; review 3-hour batch timing
- Increase supervision and quality checks during evening shifts

3

General Process Enhancements

- Implement tighter controls on ingredient quality and timing for MEAL002 and VEG003
- Monitor operator performance on high-defect lines and provide retraining

These actionable recommendations target the root causes identified across suppliers, materials, production lines, and shifts to sustainably reduce defective batch rates and optimize product quality at FreshHarvest Foods.

Appendix

1

Access to Power BI Reports

2

Access to Technical Report

3

Access to Root Cause Hypothesis Document



Thank You !

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